Cell cycle and mitosis

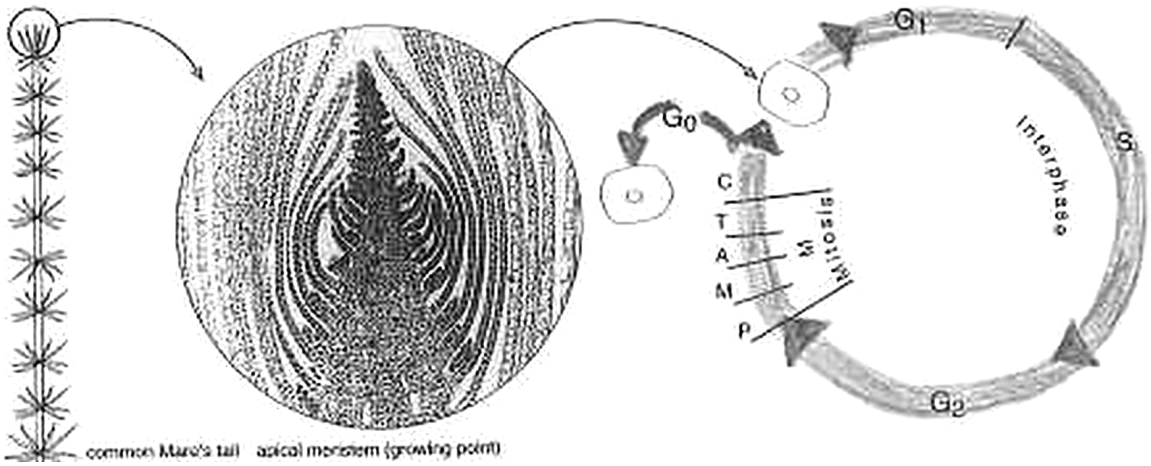


Fig. 1 Replication cycle of cells. Left: common mare's tail (*Hippuris vulgaris*) and its apical meristem (growing point).

G1 cell growth with intensive metabolism and multiplication of cell organelles

S synthesis of new DNA (replication)

G2 further growth and preparation for mitosis

M mitosis with its parts Pro- (P), Meta- (M), Ana- (A), Telophase (T)

C cytokinesis

G0 differentiation to a permanent cell

„Omnis cellula e cellula“ was the conclusion that RUDOLF VIRCHOW came to in 1855, based on his anatomical studies. His conclusion, „every cell originates from another existing cell like it“ is still valid today. WALTHER FLEMMING specified 25 years later that „every cell nucleus originates from an existing nucleus“.

Cell division takes place as long as an organism is growing or when it renews at least some of its cells. During cell division, the parent cell produces two identical daughter cells. The reproduction of cells, therefore, depends on the cells' ability to divide. Many cells that are capable of division, and thus of reproduction, can be found in the dividing tissues (meristematic tissue) of plants (figure 1), and in embryos, the skin, the epithelium of the intestine and in bone marrow. Approximately 100 million cells divide per minute in an adult human.

Following cell division, the newly formed daughter cells have the same content as the parent cell. The nucleus and cell organelles have to be divided equally between the daughter cells. Cell division thus requires the division of the nucleus (mitosis) and the even distribution of the cell organelles and the cytoplasm *(cytokinesis)*. This process is carefully prepared during the *interphase*. The proteins necessary for division are produced in the interphase (G-phases) and

the amount of chromatin doubles (S-phase). The constantly recurring sequence of interphase and cell division is called the cell cycle. It only stops when the cell starts to differentiate or die.

The events that occur during interphase can only be described by referring to methods used in molecular genetics and are therefore explained in the chapter about genetics. However, we can easily see cell division if we use an adequate light microscope (fig. 2).



Fig. 2 Stages of mitosis

Mitosis is a continuous process but in the following description it has been broken down into four stages to make it easier to understand.

Prophase

During prophase, chromatin is condensed to its transport form and, as a result, the chromosomes slowly become visible under the light microscope. The mitotic spindle, which is made up of many microtubules, starts to form between the two cell poles. The nuclear envelope fragments and the chromosomes move to the metaphase plate during *prometaphase*.

Metaphase

The mitotic spindle is complete and the chromosomes are arranged on the

metaphase or equatorial plate. We can now distinguish single chromosomes. Each cell consists of the genetic material of the mother and father. That is why there are two sets of chromosomes (2n). Each chromosome is made up of two identical sister chromatids (chromatids = 2C) because the chromosome has already duplicated itself during interphase. The two sister chromatids of a chromosome are still joined at the centromere.

Anaphase

The chromatids of a chromosome are separated from one another and connected to microtubules of the mitotic spindle. The mitotic spindle moves the detached sister chromatids to opposite poles. By the end of anaphase, one sister chromatid of each chromosome is located at each pole.

Telophase

The mitotic spindle is broken down and the chromosomes become less condensed until we can only see the chromatin. The nucleoli and the nuclear membrane reassemble.

Mitosis is followed by *cytokinesis*. Animal cells form a cleavage furrow, whereas plant cells form a growing cell plate that separates the two daughter cells.

The fate of the cell is decided after cell division. If it loses its ability to divide, it goes into the so-called G0-phase and differentiates into a permanent tissue cell. Otherwise, it goes on to a new cell cycle that again will lead to cell division.

Figure: the stages (micrograph + drawing)

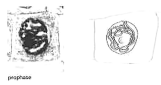
Right-hand values: sets of chromosomes (n) / sets of chromatids (C).

Interphase — 2C



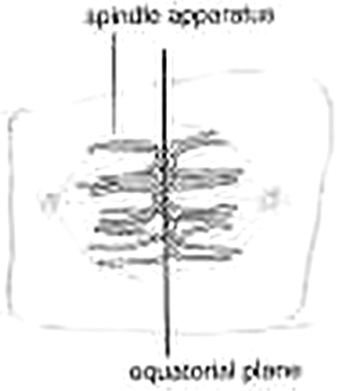


Prophase — 2C



Metaphase — 2n, 2C

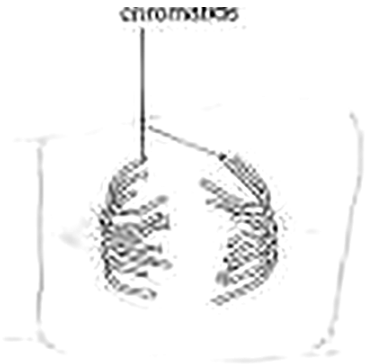




Anaphase — 1C



anaphase



Telophase — 1C



telophase

